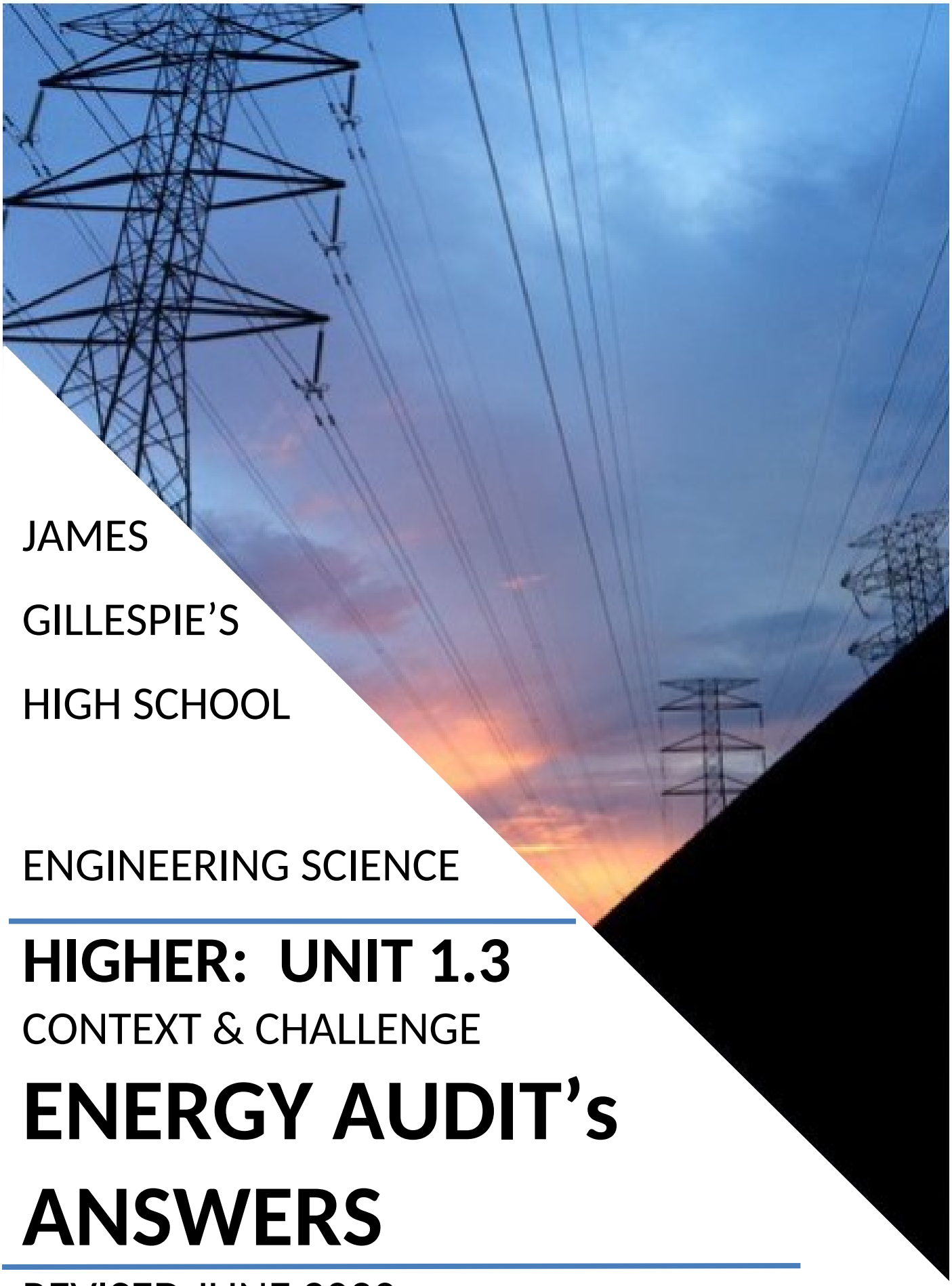


JAMES

GILLESPIE'S

HIGH SCHOOL

ENGINEERING SCIENCE

HIGHER: UNIT 1.3

CONTEXT & CHALLENGE

ENERGY AUDIT's ANSWERS

REVISED JUNE 2020

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Outcomes and Assessment standards

There are main 3 outcomes in this unit and it has been divided into three booklets. By the end of this third booklet, you will have completed outcomes 1.5. You will investigate engineering systems, problems and solutions, involving some existing and emerging technologies and consider implications relating to the environment, sustainable developments, and to economic and social issues.

This assessment applies to the following Outcomes and Assessment Standards of the *Engineering Contexts and Challenges (Higher)* Unit.

The Context and Challenge unit Outcomes are:

1. Investigate engineered objects.
2. Investigate engineered challenges and relate these too key engineered concepts.
3. Describe some aspects of the impact of engineering.

Outcome 1

The candidate will:

1.1 Analysing the needs being met by the system.
1.2 Identifying sub-systems, and describing the function of each and how they interact.
1.3 Producing system and sub-system diagrams.
1.4 Explaining the role of feedback in the system.
1.5 Carrying out an energy audit of the system.

Outcome 2

The candidate will:

2 Investigate engineering challenges and relate them to key engineering concepts by:

2.1 Identifying and describing how several different branches of engineering contribute to solving an engineering challenge.
2.2 Describing examples of the varied roles of engineers in designing, implementing, testing and controlling complex systems.
2.3 Modelling some aspect (related to one branch of engineering) of a solution to an engineering challenge.
2.4 Explaining how emerging technologies may provide improved solutions to engineering challenges.

Outcome 3

The candidate will:

3 Describe some aspects of the impact of engineering by:

3.1 Describing clearly the nature of the problem.
3.2 Describing some social and economic impacts of the solution.
3.3 Describing clearly some environmental impacts of the solution.
3.4 Identifying and describing emerging technologies which may impact future developments.

Learning Intentions

- To learn about energy forms.
- To learn about basic energy transformations.

Success Criteria

calculations related to energy audits:

— inputs
— outputs
— energy losses
— efficiency
applied calculations involving efficiency, work done and power, using:

Ew = Fd P = E/t
Ek = ½ mv^2
Ep = mgh
Ee = VIt
Eh = cmΔT
Efficiency η = Eout/Ein = Pout/Pin

Use traffic lighting to record your progress. This will also help with revision as you will be able to see what areas you need to work on. Your teacher will sign off this page after speaking to you about your self-evaluation:

Red: I need to work on this topic further to understand it.
Amber: I need a bit of help to fully understand this topic.
Green: I am confident in my understanding of this topic.

☐ I can identify basic energy transformations.

Rounding and significant figures

When answering calculations on this course, make sure you are using significant figures within your final answer. In a number, all figures are significant, except zeros at the front (to show decimal points) or at the end. For example, 1234 has four significant figures. 0.012 has only two significant figures, as does 1200. You should also take care to round your numerical answers to an appropriate number of significant figures.

$3.6 \times 10^{3} = 3600.00$ $3.6 \times 10^{-3} = 0.0036$

Relationships from Data Booklet

These are the formulae you will be provided with throughout the course and assessments. While you do not need to memorise them, you should be able to recall them and state what each means.

Energy and power

Potential energy	$E_p = mgh$	$g = 9{\cdot}8\ ms^{-2}$ (to 2 significant figures)
Kinetic energy	$E_k = \frac{1}{2}mv^2$	
Heat energy	$E_h = cm\Delta T$	$c_{water} = 4180 Jkg^{-1}K^{-1}$
Electrical energy	$E_e = VIt$	
Work done	$E_w = Fd$	
Power	$P = \frac{E}{t}$	
Electrical power	$P = VI = \frac{V^2}{R} = I^2R$	
Mechanical power	$P = Fv$	$P = 2\pi nT$ (n= no of revs per second)
Efficiency	$\eta = \frac{Energy_{out}}{Energy_{in}} = \frac{Power_{out}}{Power_{in}}$	

Prefixes and Scientific notation

Throughout this course, you should be able to recognise the different prefixes to units and be able to covert to the standard form of the unit for use in formulae. You can do this either by using scientific notation or by multiplying or dividing by the correct factor.

prefix	*symbol*	*notation*	*operation*
nano	n	$x\ 10^{-9}$	÷ 1 000 000 000
micro		$x\ 10^{-6}$	÷ 1 000 000
milli	m	$x\ 10^{-3}$	÷ 1000
kilo	k	$x\ 10^{3}$	x 1000
mega	M	$x\ 10^{6}$	x 1000 000
giga	G	$x\ 10^{9}$	x 1000 000 000

Energy

Energy is all around us and comes in many different forms. Although it cannot be seen or touched you can be sure when anything happens energy is responsible. Energy is often defined as the ability to make things happen e.g. we can give a football the energy to move by kicking it.



All energy ultimately comes from the Sun: the food we eat, the petrol used in cars, the hydrogen in rocket fuel. Some energy such as coal and oil was stored in the earth many years ago, other energy is still reaching the earth today in the form of sunlight.

If we could capture only 1% of the solar energy falling on the Earth we would have more than sufficient power for all our needs.

Work done and Energy

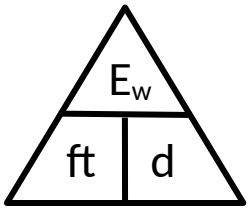
When a force is used to move an object, 'work' is said to be done. An example of this is pushing a car from position A to position B.



The amount of work you will do depends on how difficult the car is to push (the size of the force) and how far it has to be pushed (the distance). The amount of work that has to be done can be calculated using the formula:

$$W = F \times D$$

$$E_w = Fd$$

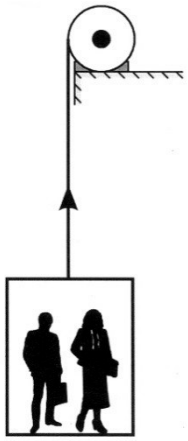


Energy Work done = Force applied x Distance moved

- Force (F)is measured in Newtons (N)
- Distance (d) is measured in metres (m)
- Energy Work done (W or E_w) is measured in Joules (J)

Teacher worked example

A lift raises a mass of 1000kg to the next floor in the Union Square shopping centre, which is 20m up. Calculate the minimum amount of work that must be done by the winch.



Remember – Gravitational field strength is always 9.81 N/kg. The gravitational field strength is provided in the data booklet as 9.8 ms^{-2} (don't worry about the units as it is the same).

Convert the weight of lift into a force = mg (mass x gravitational field strength)

(What are your knowns?) m = 1000kg and g = 9.8 N/kg d = 20m

Force = 1000 x 9.8

= 9800 N

F = 9800N, d = 20m

E_w = force x distance

= 9800 x 20

= **196,000 J**

Work done questions

1. In lifting an engine out of a car, a mechanic uses a mechanical device called a block and tackle. How much work is done if a force of 500 N is used in pulling the rope a distance of 4.2m?

(What are your knowns?) F = 500N, d = 4.2m

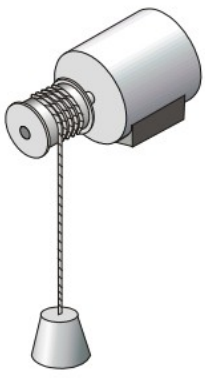
Ew = Fd
Ew = 500 x 4.2
Ew = 2100J

2. During the loading process a fork lift truck lifts a pallet of bricks of mass 740 kg up to a height of 2.2 m. Calculate the amount of work done to move the pallet.

(What are your knowns?) F = mg g = 9.8N/kg m = 740kg d= 2.2m

F = 740 x 9.8 = 7252N
Ew = Fd
Ew = 7252 x 2.2
Ew = 16000J

3. A weight lifter "snatches" a dumbbell from the floor to arm's length, a vertical height of 1.4 m. Calculate the mass in kg of the dumbbell if the minimum



work done in lifting it is 1800J.

(What are your knowns?) Ew= Fd Ew= 1800J d = 1.4m
F = Ew ÷ d
F= 1800 ÷ 1.4
F = 1285.71N F= mg
m= F ÷ g
m = 1285.71 ÷ 9.8
m = 131kg (3s.f.)

4. A mass of 50 kg is raised to a height of 5.6 m by a rope, which is wound around a motor shaft of diameter of diameter 150 mm as shown. Determine the amount of work done by the motor and the number of revolutions made during the lift.

(What are your knowns?) d = 5.6m = 5600mm m = 50kg g = 9.8N/kg Diameter = 150mm
F = mg F = 50 x 9.8 = 490N
Ew = Fd
Ew = 490 x 5.6
Ew = 2744J
Circumference = 2 or (in data booklet)
Circumference = 2 x x 75 = 471mm
Number of revolutions = Distance moved / Circumference
5600 / 471 = 11.9 revolutions

Energy Types

In addition to Work Done, energy is defined as the capacity of a body to make things happen (or to work). It is measured in Joules (J). We know that to do more work we require more energy but we can use various types of energy to do this work for us. Energy comes in several different forms.

Kinetic Energy

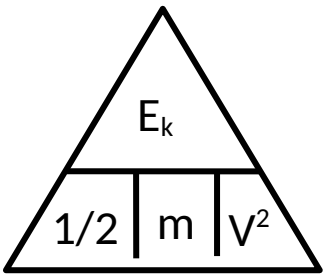
Kinetic energy is the energy of movement. It is the name given to the energy a body possesses due to its motion. The person below can be described as having kinetic energy because they are moving.



The kinetic energy of a moving object is dependent on two factors; the mass (m) in kg of the object and its velocity (v) in m/s.

Kinetic energy is calculated using the formula:

$$E_K = \frac{1}{2}mv^2$$



Kinetic energy = ½ x mass x velocity²

- Mass (m) of the object is measured in kg.
- Velocity (v) is measured in m/s or ms^{-1} (either one is accepted as they are the same unit of measurement).

Teacher worked example

The total mass of a go-kart and the driver is 90kg. The go-kart is travelling at 40 m/s, how much kinetic energy does it possess? (What are your knowns?) m= 90kg V = 40m/s

E = ½ mv²

= ½ x 90 x 40²

= ½ x 90 x 1600

= 72,000 joules

= 72 kJ

Potential Energy

Potential energy can be best thought of as energy stored in a static object. It can be due to how high the object is above a datum (starting point), or due to the fact that work has already been done on the object and the energy is stored in it. The bucket supported by the pulley in the diagram below contains potential energy.

The formula most commonly used to calculate potential energy is:



$$E_P = mgh$$

Potential energy = mass x gravity x height

- Mass (m) of the object and is measured in kilograms (kg)
- Height (h) of the object above the ground or datum is measured in metres (m)
- Gravitational field strength (g) acting on the object and is always 9.8N/kg or ms^{-2} on Earth.

Teacher worked example

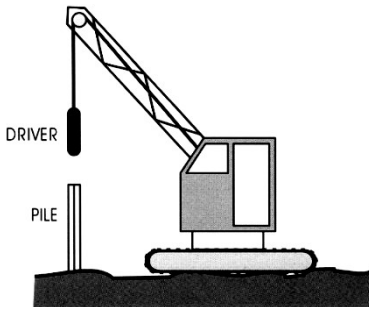
Metal piles are driven into the ground using a pile driver. This consists of a 500kg driver, which is raised by a winch to a height of 3m, then released. Calculate the potential energy stored when the driver is at its maximum height. (What are your knowns?) m = 500kg h = 3m g = 9.8N/kg

E_p = mgh

= 500 x 9.8 x 3

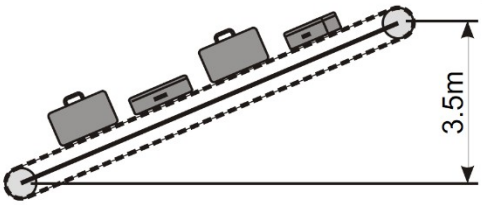
= 14, 700 Joules

= **14.7kJ**



Potential energy questions

5. Baggage handlers at an airport, place suitcases onto a conveyor belt, which lifts them up to the hold of the aeroplane as shown. Calculate the potential energy stored when a 20 kg case is moved up the belt.



(What are your knowns?)

Ep =mgh m = 20kg g = 9.8N/kg h = 3.5m

Ep = 20 x 9.8 x 3.5

Ep = 690J

6. At what height must a drum of mass 100 kg be suspended above the ground if it possesses 4 kJ of potential energy?

(What are your knowns?) Ep = mgh Ep = 4000J m = 100kg g= 9.8N/kg
h = Ep ÷ mg
h = 4000 ÷ (100 x 9.8)
h = 4.1m

7. Calculate the potential energy available from a reservoir holding 1800 litres of water at a height of 260 m. (1 litre of water = 1 kg)

(What are your knowns?) m = 1800kg g = 9.8 h = 260m
Ep = mgh = 1800 x 9.8 x 260 = 4.6MJ

8. Metal piles are driven into the ground using a pile driver. This consists of a driver being raised to a height of 5.1 m above the ground and then released. Calculate the weight of the driver if the potential energy stored when it has been lifted is 9,810 Joules.

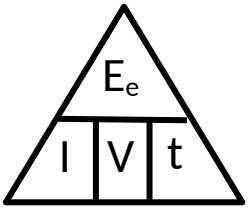
(What are your knowns?) w = mg Ep = 9810J h = 5.1m
w = Ep ÷ h
w = 9810 ÷ 5.1
w = 1.9KN

Electrical Energy, Ee

Electrical energy is one of the most convenient and commonly used forms of energy since it can be transported easily from place to place (along electrical cables) and can be easily changed into other forms of energy.

The formula for calculation electrical energy is:

$$E_e = ItV$$



Electrical Energy = Current x Voltage x time

- Current (I) is measured in Amps (A)
- Voltage is measured in Volts (V)
- Time is measured in seconds (t)

Teacher worked example

An electric hob in home economics has an operating voltage of 230 volts with a current of 5 amps. Calculate how much electrical energy has been used if the hob takes 5 minutes to heat up a pot of soup.

Remember to always convert your time into seconds.
Therefore 5 minutes x 60 = 300 seconds.
(What are your knowns?) I = 5A V = 230V t = 300 seconds

E_e = IVt
= 5 x 230 x 300
= 345,000 joules
= 345 kJ

Electrical energy questions

9. A hot air hand drier is activated for 30 seconds when the switch is pressed. The drier operates from a 230 V supply and draws a current of 12 A. Calculate the amount of electrical energy used when the drier is operating.

Ee = Itv (What are your knowns?) I = 12A t = 30s V = 230V
Ee = 120 x 30 x 230
Ee = 828 000J = 82.8KJ

10. If the amount of electrical energy used by a 110 V, 30 A dc motor is 1.98 MJ. How long has the motor been in operation for?

Ee = Itv (What are your knowns?) Ee = 1.98MJ = 1 980 000J I = 30A V = 110V
t = Ee ÷ Iv
t = 1 980 000 ÷ (30 x 110)
t = 1 980 000 ÷ 3300
t = 6000s

11. A 3 Amp turbine is used to drive a 110 V electrical generator. How much electrical energy does the generator produce in one hour?
(What are your knowns?) I = 3A t = 3600s V = 110V
Ee = Itv
Ee = 3 x 3600 x 110
Ee = 1.2MJ

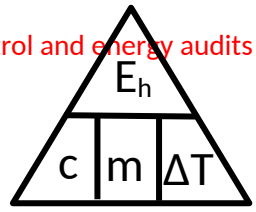
Thermal Energy



Thermal energy is the energy transferred to a body, which results in a change in the body's temperature. The diagram below shows a kettle boiling, therefore a certain amount of thermal energy was required to raise the temperature of the water in the kettle to boiling point.

Thermal energy can be calculated using the formula:

$$E_h = mc\Delta T$$



Heat Energy = heat capacity of material x mass x change in temp

- Mass (m) is measured in kg
- Specific heat capacity (c) is measured in Joules per Kilogram per Kelvin (or Celcius) (J/kgK or J/kg°C)
- Change in temperature (ΔT) ('Delta T') is measured in Celsius (°C) or Kelvin (K)

Important: The specific heat capacity of a substance, is the amount of energy required to raise the temperature of 1kg of the material by 1 degree celsius.

Teacher worked example:

A hot water tank contains 200 litres of water at 18°C. Calculate how much energy is required to raise the temperature of the water to 50°C. 1 litre of water has a mass of 1kg. The specific heat capacity of water is 4180 J/kgK. The specific heat capacity of a liquid changes depending on the type of liquid. The provided data booklet will specify what value to use.

C_{water} = 4.180 KJ/KgK or 4180 J/KgK (C_{water} = 4200 J/KgK in the Higher data book)

Change in temperature (ΔT) = T2 – T1

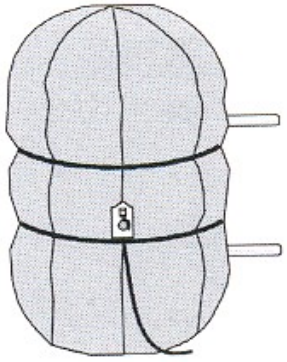
= 50 – 18 = 32°C

E_h = cmΔT

= 4180 x 200 x 32

= 26752000 J

= 26.752 MJ

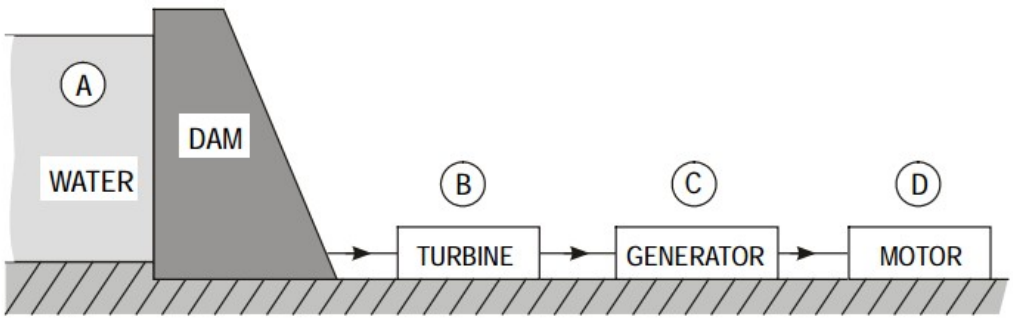


Where:

- **m** is the mass of the material in kg.
- **ΔT** is the change in temperature in degrees (Celsius or Kelvin).
- **c** is the specific heat capacity of the material being heated.

Mixture of energy questions

12. Name the energy output at each of the stages A, B, C and D.



A = Potential
B = Kinetic

C = Electrical
D = Kinetic

13. A race car drives around a circuit at a speed of 50 m/s. If the car has a kinetic energy of 500 kJ what is the mass of the car?

(What are your knowns?) V = 50m/s Ek = 500KJ = 500 000J m = ?

$Ek = ½mv^2$
$m = Ek ÷ ½v^2$
$m = 500\ 000 / 0.5 \times 50^2$
m = 400kg

14. A girl with a mass of 50 kg is riding on her bicycle and has a kinetic energy of 2.5 kJ. What speed is the girl moving at and what is the kinetic energy of the bicycle if it has a mass of 30 kg?

(What are your knowns?) $Ek = ½mv^2$ m = 50kg Ek = 2.5KJ = 2500J m = 50kg v = ?
$V^2 = Ek ÷ ½m$
$V^2 = 2500 ÷ (0.5 \times 50)$
$V^2 = 2500 ÷ 25$
$V^2 = 100$ 100 = 10 m/s b) $Ek = ½mv^2$ $½ \times 30 \times 10^2 = 150$ ½ x 30 x 100 = 1500J

15. Calculate the potential energy available from a reservoir tank holding 1800 litres of water at a height of 26 m. (1 litre of water = 1 kg)

(What are your knowns?)

Ep = mgh g = 9.8m/s h = 26m m = 1800kg
Ep = 1800 x 9.8 x 26 Ep = 458640J = 459 000J

16. A fairground roller coaster consists of many high and low points on it's track. If the highest point at the beginning of the ride is at a height of 50 m and the height at the end is 5 m, what is the change in potential energy between the start and end for a person of mass 80 kg?

(What are your knowns?)
Ep = mgh D = 50-5 = 45m m = 50kg
Ep = 80 x 9.8 x 45
Ep = 36 000J

17. A portable electrical generator can deliver energy at a rate of 6 kJ per second. Calculate the current that can be drawn from the generator if the electricity is supplied at a voltage of 110 Volts.

(What are your knowns?)

Ee = Ivt Ee = 6KJ = 6000J/sec V = 110v I = ?
I = Ee ÷ vt
I = 6000 ÷ 110
I = 54.54A

18. A 3 Amp turbine is used to drive a 110 V electrical generator. How much electrical energy does the generator produce in one hour?

(What are your knowns?) I = 3A v = 110 t = 1 hour = 60 minutes = 3600seconds

Ee = Ivt = 3 x 110 x 3600
Ee = 1 188 000J = 1 190 000J

19. A sport hall has dimensions of 8m x 30m x 10m. Assuming no heat losses, calculate the amount of heat energy required to raise the temperature of the air in the hall from 10°C to 20°C. The density of air is approximately 1.3 kg/m^3 and the specific heat capacity is 850 J/kgK.

(What are your knowns?) Volume = 2400m^3

E_h = cmΔT m 2400 x 1.3 = 3120kg ΔT = 10°C
Eh = 850 x 3120 x 10
Eh = 26 520 000 = 26 500 000J

20. 57 kJ of thermal energy are supplied to 1.7 kg of oil having a specific heat capacity of 2.7 kJ/kgK. If the initial temperature of the oil is 3°C, what will be its final temperature?

(What are your knowns?) Eh = 57 000J m = 1.7kg c = 2.7KJ/kgK = 2700J/kgK

E_h = cmΔT
T = Eh ÷ cm
T = 57 000 ÷ (1.7 x 2700)
T = 57 000 ÷ 4590
T = 12.418 + 3°C = 15.4°C

21. Calculate the heat energy required to heat 2 kg of water from a temperature of 20°C until it begins to boil.

E_h = cmΔT T = 20°C Boil = 100°C ΔT = 80°C m = 2kg c = 4200 J/kgK
Eh = 4200 x 2 x 80
Eh = 672 000J

Calculating Power

Power is simply a measure of how fast energy is used or transferred. Whenever there is current in a wire, electrical energy in transformed into another form. This could be heat, light, kinetic energy etc, depending on the electrical appliance being used. Sometimes these transformations are intentional (light) and sometimes they are not (loss energy as heat). Power is a measure of the rate of energy transfer in one second, i.e. the energy transferred in one second. It gives an indication of how quickly the energy is changed from one form to another. Power in measured in joules per second or watts (W).

$$P = \frac{E}{t}$$

Power = Energy transfer ÷ time

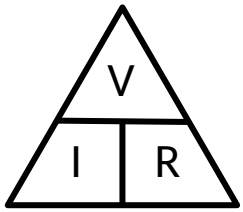
The Energy used in this equation can be in <u>any form</u>, from work done (J), kinetic (J), electrical (J) or indeed any type of energy - providing it is measured in joules.

- Power (P) is measured in watts (W). Either unit is acceptable as P is equivalent to one joule per second (Js^{-1})
- Energy transfer (E) is measured in joules (J)
- Time (t) is measured in seconds (s)

Resistance formula using Ohm's law

$$V = IR$$

V = potential difference (V)
I = current (A)
R = resistance (W)



To calculate Energy (E) = P x t, power (P) can be replaced with the value of IV from Ohms law as shown:

Energy (E) = P x t

Therefore Energy (E) = IV x t The Power formulae however is actual written as **P = IV**

Power Relationships (Formulae)

In addition to the definition for power above, there are three other relationships which allow us to calculate the power of an electrical component or appliance. These formulae relate power to potential difference (voltage), current and resistance.

$$P = IV$$

P = power (watts, W)
I = current (amperes, A)
V = potential difference (volts, V)

$$P = \frac{V^2}{R}$$

P = power (watts, W)
V = potential difference (volts, V)
R = resistance (ohms, Ω)

$$P = I^2 R$$

P = power (watts, W)
I = current (amperes, A)
R = resistance (ohms, Ω)

Teacher worked example

If an electric light bulb uses 60 kJ of energy in 10 minutes, what is the power rating of the bulb?



E = 60,000 joules

t = 10 minutes x 60 seconds, therefore t = 600s

$$P = \frac{60\,000}{600}$$

P = 100 Watts

Power questions

22. Calculate the power of a pumping engine which must pump water from the bottom of an excavation 3 m deep to ground level, if the water gathers at a rate of 220 litres every minute. (1 litre of water has a mass of 1 kg.)

Calculate energy first: Ew = Fd, F = mg F = 220 x 9.8 = 2156N d= 3m

= 2156 ÷ t

= 2156 ÷ 60 Seconds

P = 35.93 Watts

23. A portable crane is used to lay sections of a sewage drain in an excavation 4 m deep. Each section has a mass of 800 kg. If the average time taken to lay one section is 3 min, calculate the power required if the crane must raise the section 2 m prior to lowering it into the excavation.

24. A press used to punch a tunnel section into position is to operate by oil pressure and is to exert a force of 2 MN. Calculate the power developed if the ram of the press is to operate once every 40 min and moves 240 mm in this time. If the diameter of the press ram is 150 mm, calculate the oil pressure needed to exert the 2 MN force.

25. 20 containers of average weight 180 kN per container is to be loaded on to a bulk carrier. Each container has to be raised 15 m by the loading crane, move over the ship and lowered 18 m. Calculate:



i) The total work done in loading the bulk carrier.

ii) The power developed by the crane if the total time taken to load one container is 5 min.

Conservation of Energy

In everyday usage, energy conservation means using less energy to do the same work. Examples of this include improving the heat insulation of buildings and homes, making cars which use fuel more efficiently… etc. In Technological Studies terms 'conservation of energy' has a different meaning. Instead it is looked upon as a rule. The Rule states that energy cannot be created or destroyed, but can only be changed from one form to another. It is 'transformed' or 'converted'. This rule is also termed a natural law.

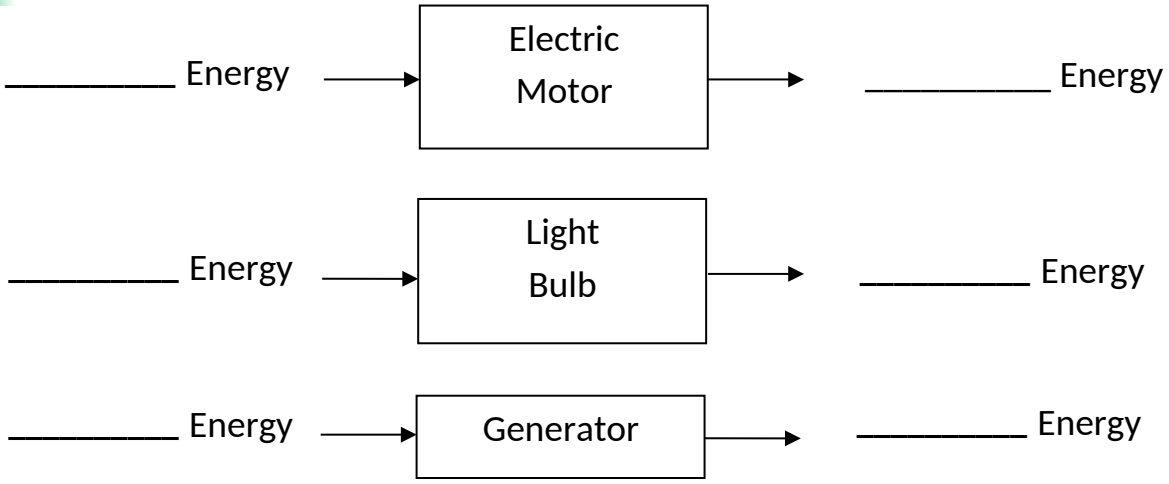
The law of conservation of energy

The law of 'conservation of energy' states that for a closed system, where no energy goes in or out, the total energy within the system must always stay the same, although the form of energy may change. On the other hand, in an open system such as a power station, this rule leads to the conclusion that the total energy input to the system must be exactly equal to the total energy output. The extent to which the output energy is able to do useful work (of the desired type) is called the efficiency of the system. We can calculate this by comparing the useful output form the system with its energy input.

Energy Transformations

How energy can be transformed or converted is very important to a technologist. Some forms of energy is directly interchangeable (e.g. potential to kinetic), but others need to go through several changed to arrive to the desired form.

26. Copy and complete these diagrams of simple transformations.



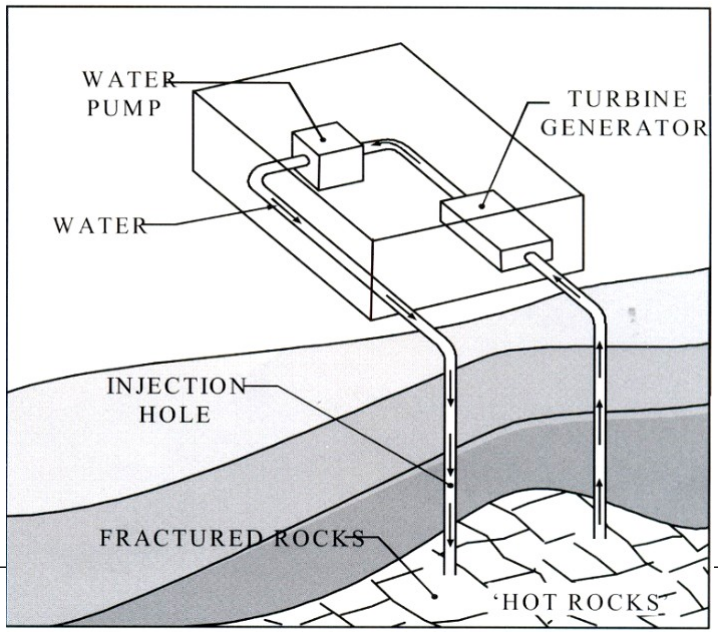
More energy transformations

Some systems will require more than one energy conversion to get the desired energy output. An example of this is converting geothermal energy into electrical energy.

Geothermal Energy

The earth is a massive reservoir of natural heat energy. At its birth the earth was a molten mass at a temperate of 6000°C approx. The earth has been cooling down ever since and will continue to do so for millions of years. In some places, the natural heat energy reaches close to the surface. Occasionally the molten materials cause the heat escape and this creates volcanoes. It is also known to heat up underwater reservoirs to create hot springs and geysers.

If this geothermal energy could be harnessed at temperatures of 250°C or higher, it could be used to create electricity, the heat being used to turn water into high-pressure steam to drive turbine generators. To achieve these temperatures however it would be necessary to drill deep into the earth's surface to reach "hot-rocks". In Britain it would be necessary to drill as deep as 6000 metres (4 miles approximately) to reach useful temperatures.



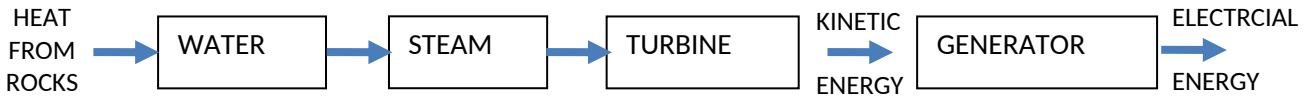
This is one method of collecting geothermal energy for transformation into electricity. This system consists of 2 boreholes, which penetrate the Earths

crust and bore into a region of hot rocks. During construction an explosive charge would be detonated at the bottom of the injection hole to fracture the rocks.

When operating, water would be forced down this hole under pressure. It would then penetrate the hot rocks, pick up the heat, and return to the surface through a second borehole as steam. This stream would then be released through a pressurised system to drive the turbine generators.

The energy conversion could be described as follows:

The heat energy in the rocks is transferred to the water in the injection hole pipe. This hot water is changed to high-pressure steam and transported to the surface. The heat energy from the steam is used to turn the blades of a turbine, producing rotational kinetic energy, which is used to create electricity from a generator.



Energy conversion questions

27. Copy and complete the following system diagrams showing the energy conversions taking place.

a) A stretched elastic band

Potential to kinetic

b) Electric kettle

Electrical to Thermal

c) A wind-up toy car

Potential to Kinetic

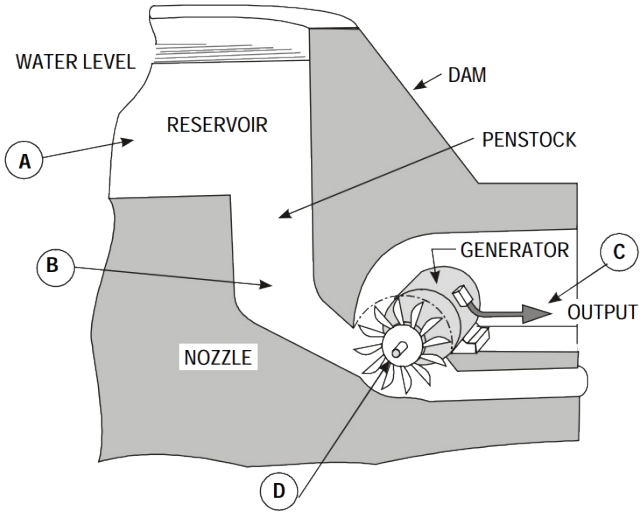
d) Water passing over a waterfall

Kinetic

Energy Conversion

It is important to be able to identify where energy is converted from one form to another in a system.

28. A simplified hydroelectric power station is shown as an outline diagram.

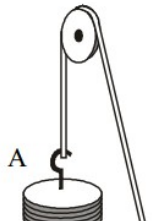


a) State the form of energy at each of the points A, B, C and D.

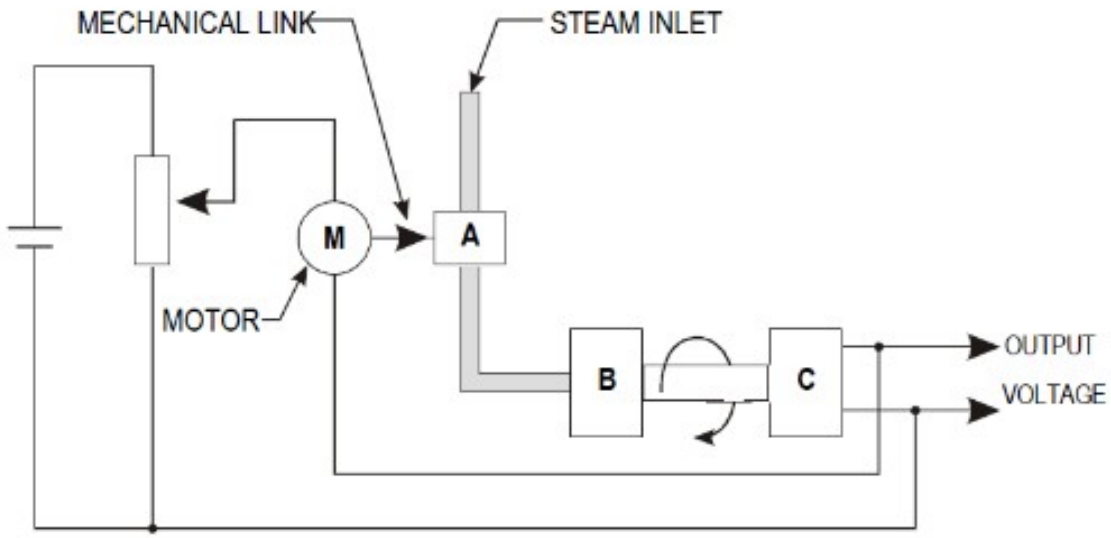
b) Complete the following statement to describe, using appropriate terminology, the energy changes, which take place between stages A and D.

Example: At A the water has ..... energy, but as the water flows down the penstock

29. Identify the energy types at points A – H and describe the energy changes in the system below.



30. A diagrammatic representation of a simplified Thermal Power Station is shown below.



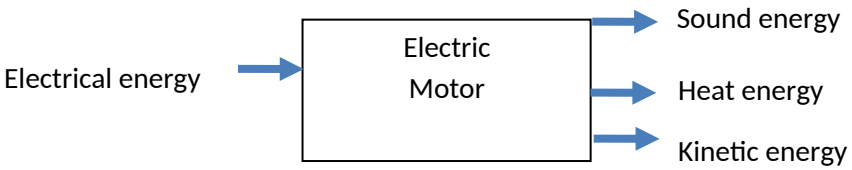
a) List the energy changes that take place between the input and the output of the Power Station.

Energy 'Losses' during transformation

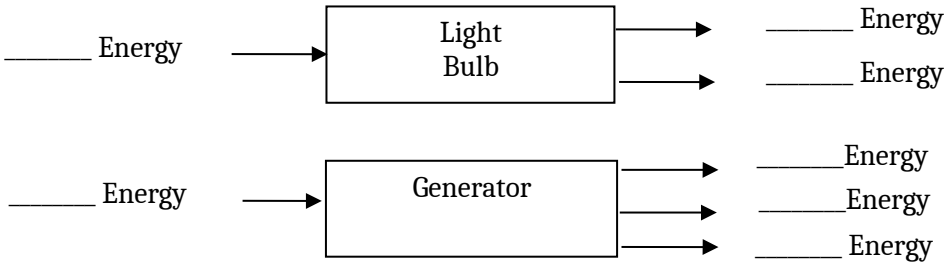
Although we have stated that energy cannot be destroyed, and that the energy output from a system is equal to its input, not all the energy is used efficiently. When an energy conversion takes place there is always an energy change that we do not desire – usually in the form of heat sound or friction from the moving parts of the mechanism. If we look at the simple energy conversions from earlier we can expand these to show the waste energy, or energy losses.

Teacher worked example

An electric motor converts electrical energy to kinetic energy, but while doing so also creates sound energy and heat energy.



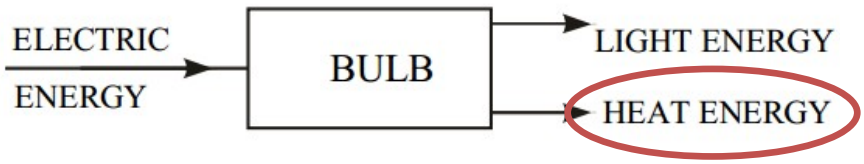
31. Complete these diagrams showing all the energy losses.



Energy Losses

Although we have stated that energy cannot be destroyed and that the energy output from a system is equal to the energy input to the system, not all the energy in the system is used efficiently.

When an energy conversion takes place there is always an energy change that we do not desire - usually a loss in the form of heat, sound or friction from moving parts of a mechanism.



Energy transformation questions

32. The diagram below shows the subsystems in a power station.



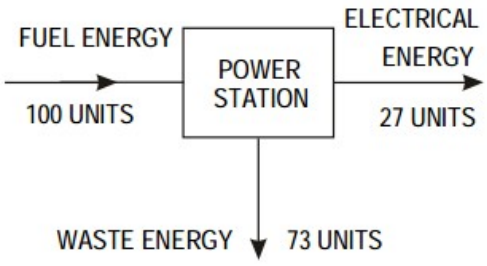
A) Use the key to identify the fluid at points A, B, C, D, and E.

B) Describe how the power station operates.

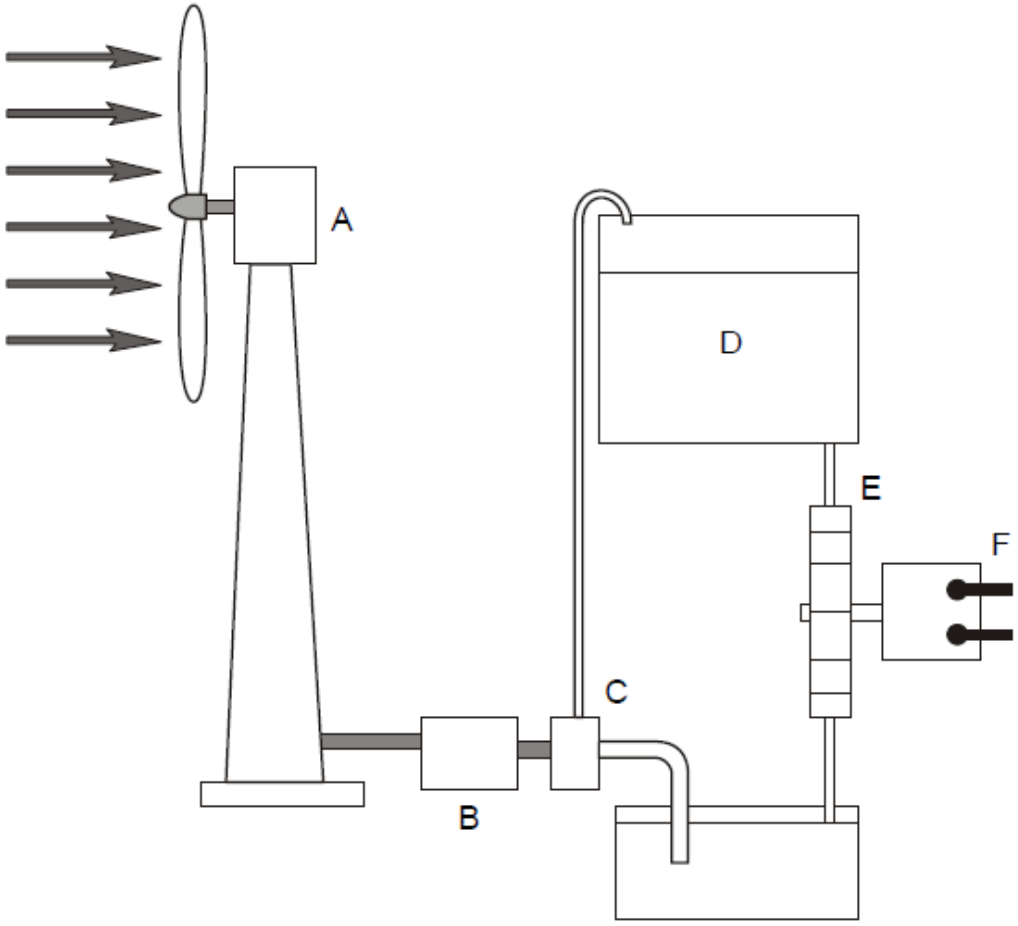
C) What is the purpose of the heat exchangers?

D) Where is energy lost in the system?

E) From the information given below, calculate the overall efficiency of the power station.
Efficiency = Electrical Energy / Fuel Energy x 100%



33. The diagram below shows a method of using the energy from wind. Identify the forms of energy at points:
A (wind vane), B (generator), C (pump), D (water tank), E (water wheel) and F (generator).



b) Describe the energy changes taking place within the system.

c) Identify and describe any energy losses within the system at points A, B, C, D, E and F.

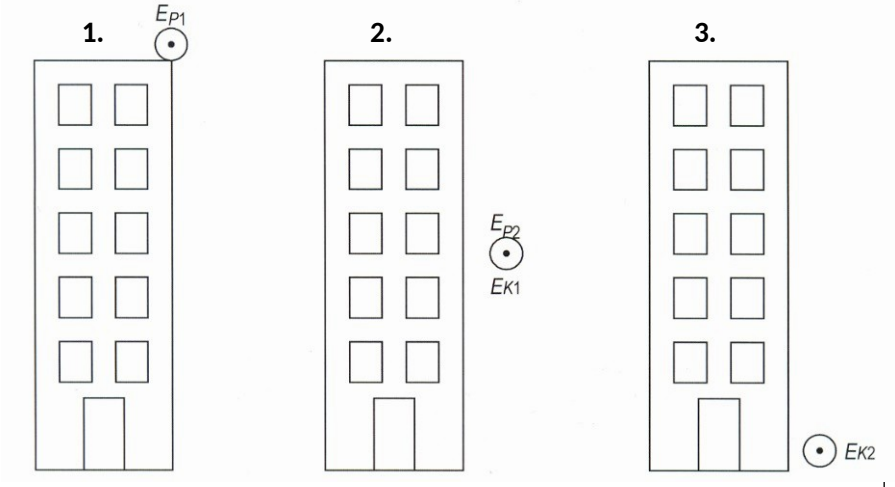
Calculating energy transfers

It has already been stated that energy cannot be created or destroyed – it can only be transformed to another energy. During an energy transformation, the total energy contained within any closed system remains constant. Knowing the total amount of energy at the start (or end) of any energy transformation tells us the total energy at any given time during the transformation.

Worked example

1. Consider a ball that is released from a high building. When the ball is at the top of the building, it has gravitational potential energy (E_{P1})

The total energy, $E_T = E_{P1}$



2. When the ball is released, it falls and some of the potential energy (E_{P2}) is converted into kinetic energy (E_{K1}). The total amount of energy (E_T) the ball possesses at this time is equal to the potential energy plus the kinetic energy.

The total energy $E_T = (E_{P2} + E_{K1}) = E_{P1}$

3. Just as the ball hits the ground, it no longer has any potential energy – all the potential energy (E_{P1}) has been converted into kinetic energy (E_{K1}). The total amount of energy now consists of the kinetic energy alone.

$$\text{Total energy } E_T = E_{K1} = (E_{P2} + E_{K1}) = E_{P1}$$

Teacher worked example: transfer of energy (potential to kinetic)

A body of mass 30kg falls freely from a height of 20 metres. Find its final velocity and kinetic energy at impact. First calculate the initial Potential Energy (E_p).

M = 30kg, H = 20m, g = 9.8N/kg

$E_P = mgh$
$= 30 \times 9.8 \times 20$
$= \underline{5.88\ kJ}$

This potential energy is converted or transferred into kinetic energy, which means that the kinetic energy

at impact is equal to 5.88 kJ. But this is not the end of the calculation! NO NOT APPLY 'sig figures' until the end calculation.

To calculate the final velocity of the body we begin by taking E_k = 5.88 kJ.

$E_K = ½mv^2$

m = 30 kg, v^2 = ?, Ek = 5880J

$5880 = ½ \times 30 \times v^2$

$5880 = 15 \times v^2$

$v^2 = 5880 \div 15$

$v^2 = 392$

$v =$ = 19.8 m/s

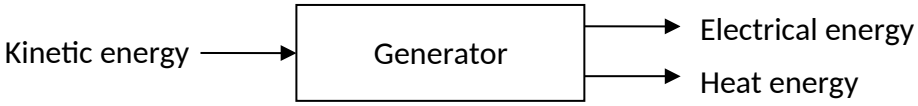
This type of calculation can be completed for any type of energy conversion. Knowing the total amount of energy at any given time (beginning/middle/end) is the same at any other given time.

Energy audits

It has already been stated that energy cannot be created or destroyed, it can only be made to change form.

During energy transformations all the energy that goes *in* the system must come *out*. It is not possible to 'destroy' energy – it must go somewhere. Not all energy going into a system comes out as useful energy – this is called **waste energy**, or ***lost energy***.

An example is a generator which is designed to convert kinetic energy into electrical energy, however, due to the frictional forces, some heat energy will also be produced. Another example of this is that a light bulb turns electric energy into light energy, but it also creates heat – heat is the lost energy. Since we know that the total energy in any closed system must be constant, we can carry out meaningful calculations if we remember to take all types of input and output energy into account. They are read in the same way as a basic systems diagram.



In the generator example above:

- the input energy is in the form of kinetic energy.
- the total output energy will be electrical energy, plus heat energy.

Hence, through conservation of energy $E_k = E_e + E_h$

In order to ensure that we have taken all energies into account, it is useful to carry out an **energy audit**. An energy audit is a list of all energies coming in and going out of a system. The total for the energies in **must** be the same as the totals for the energies out.

Efficiency

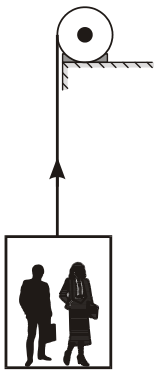
It is possible to look at how well an energy system is operating by calculating its efficiency. The efficiency of an energy transformation is a measure of how much the input energy appears as useful as the output energy. This can be calculated using the following equation:

$$\text{Efficiency} = \frac{\text{Useful energy Output}}{\text{Total energy Input}}$$

$$\eta = \frac{E_{out}}{E_{in}}$$

Note – **η** is the ratio of output to input energy. This can never be greater than 1. In order to convert efficiency (**η**) to a percentage, it has to be multiplied by 100.

Teacher worked efficiency example



An electric lift rated at 110V, 30A raises a 700kg load to a height of 20m in 2 minutes. By considering the electrical energy input and the potential energy gained by the mass, determine the percentage efficiency of this energy transformation.

Remember – translate the time into seconds! (2 mins x 60s = 120s)

The energy going into the system is electrical:

I = 30A, V = 110V, t = (2 x 60) = 120s

$E_e = IVt$

$= 30 \times 110 \times 120$

$= \underline{396\,000 J}$

The output Energy is potential so:

$E_P = mgh$

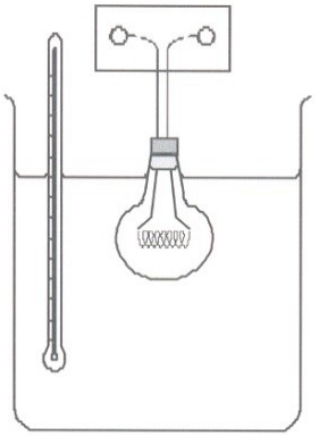
$= 700 \times 9.8 \times 20$

$= \underline{137\,200 J}$

$$\eta = \frac{E \text{ out}}{E \text{ in}} = \frac{\text{potential energy}}{\text{electrical energy}} = \frac{137\,200}{396\,000} = 0.346$$

Percentage efficiency = 0.346 x 100 = 34.6% (35 % 2 s.f)

Teacher worked efficiency audit example



In order to estimate the efficiency of an old filament electric light bulb, the bulb is immersed into a beaker of water as shown. Assuming all the heat energy is transferred to the water, it is possible to work out the efficiency of the bulb as a light producer.

Data:

Energy type going into the system is electrical (E_e)
Power supply = 12 V
Current Drawn by the bulb = 5 A
Volume of water in beaker = 0.5L
Initial temperature of water = 18°C Temperature of water after 10 minutes = 30°C
Time lightbulb is immersed for 10 minutes

Calculations:

t = 10 minutes x 60 seconds = 600s

E_e in = IVt
= 5 x 12 x 600
= 36,000 J = **36 kJ**

Energy OUT losses is heat (E_h) and light (E_L)

ΔT = 30 – 18 = 12°c

E_h out = cmΔT
= 4180 x 0.5 x 12
= 25 080 J

The specific heat capacity (c) for water is $4180 Jkg^{-1}K^{-1}$

(This is also in the data booklet)

'Energy out' **must equal** the amount of 'energy in'. However 'minus' potential energy losses.

Energy in = 36 000 J
Energy out = 25 080J

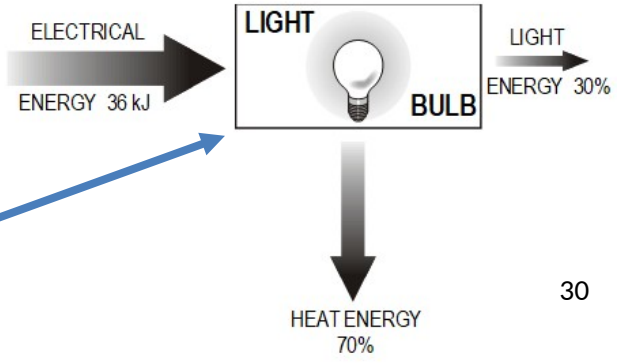
Therefore energy losses: Total energy in – energy out
Energy losses = Energy out + $E_{heat\ loss}$ + $E_{light\ loss}$
Energy losses = 36 000 – 25 080 = 10 920 J

Efficiency = Energy losses / Total energy in = 10 920 / 36 000 x 100% = **30.3%** (30% s.f.)

Waste energy (heat energy) 70% of input = 25 200J

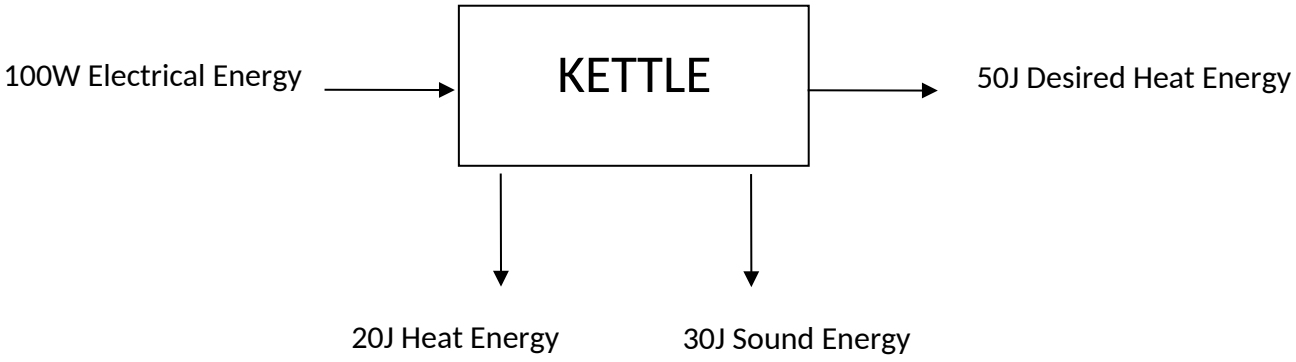
(25KJ 2s.f)

This can be represented by a systems diagram opposite.



System diagram

From the example diagram shown below, we can easily see that the system is only 50% efficient. Typically you may get asked to either draw a diagram, or interpret information from it.



From the example diagram shown below, we can easily see that the system is only 50% efficient. Typically you may get asked to either draw a diagram, or interpret information from it.

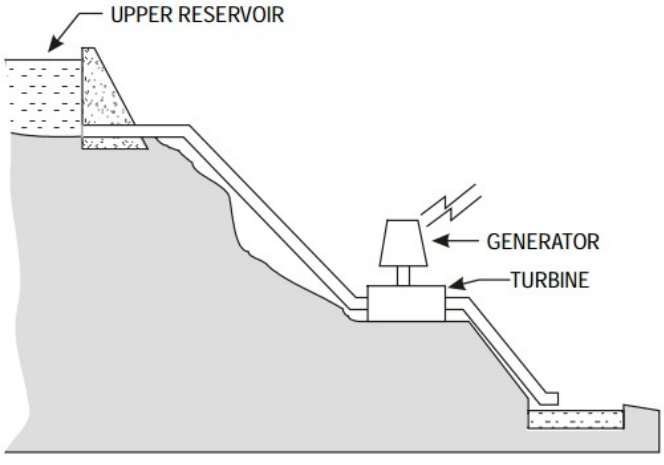
Efficiency energy questions

34. An electric kettle is rated at 240 V, 10 A. When switched on it takes 3 minutes to raise the temperature of 0.5 kg of water from 20 °C to 100 °C.

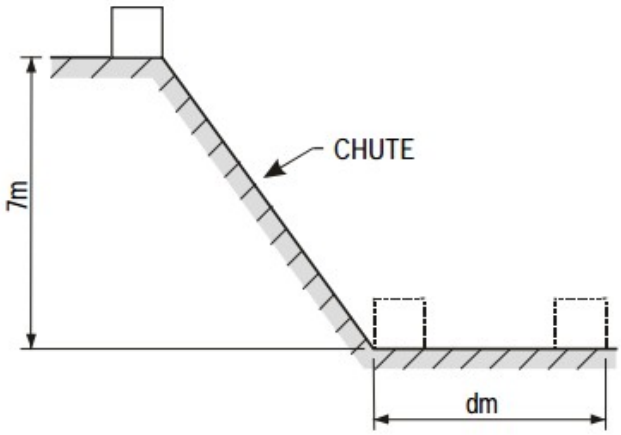
Determine :

a) The electrical energy supplied in the three minutes.
b) The heat energy required to raise the temperature of the water.
c) The efficiency of the kettle.

35. In a Hydroelectric electricity generating stations, water is allowed to flow downhill through a turbine, which is connected to a generator. The water falls through a vertical height of 500 m at a rate of 5,000 kg/s. If the energy transfer is 65% efficient, determine the amount of electrical energy produced per second.



36. Boxes in a factory are transferred from one floor to another using a chute system as shown. The boxes start from rest at the top of the chute and during the decent there is a 40% loss of energy. The boxes have a mass of 10 kg each.



i) Calculate the velocity of the boxes at the bottom of the chute.

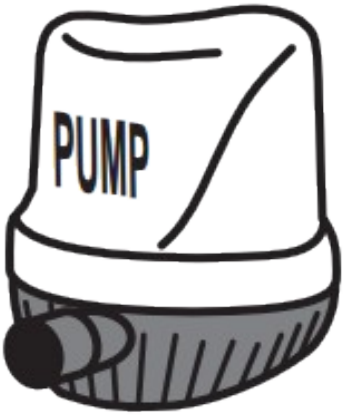
ii) What is the distance "d" that each box will travel along the bottom floor before coming to rest if the frictional force opposing the motion of each package is 25% of its weight?

37. The motorised pump shown produces the following performance data, when tested by a maintenance engineer.

Energy output: 97 J in 5 seconds

Current drawn: 3·0 A

Voltage: 12 V



Draw an energy audit diagram for the pump, showing its efficiency. 3

Energy questions A

1) A 5kg mass is raised steadily through a height of 2 m. What work is done and what is the body's potential energy relative to the start?

d = 2m g= 9.8 m = 5kg
Ew = FD F = mg F = 5 x 9.8 = Ew = 49 x 2 = 98J

2) A body of mass 30 kg is projected vertically upwards with an initial velocity of 20 m/s. What is it's initial kinetic energy and to what height will it rise?

3) A mass of 20 kg is allowed to fall freely from a certain height above a datum. When the body is 16 m above the datum, it possesses a total energy of 3531 J. What is the starting height of the object?

4) In a stamping machine, the die has a mass of 35 kg and falls through a height of 2m onto a metal block. If the depth of indentation is 10 mm, find the average stamping force. Assume no rebound.

5) A car manufacturer carries out a safety test on the bumper system of a new model as shown below. The test involves propelling the car at a constant speed of 8 km/hour towards a vertical wall. The resistance to motion of the car is 600 N and the mass of the car is 800 kg. Four life sized dummies are placed in the car to allow additional data to be recorded. Each dummy has a mass of 70 kg.

The bumper system is designed to absorb all the energy at impact. During the test, the bumper was found to have compressed by 30 mm before returning to its original shape.

Calculate:

a) The kinetic energy of the car before it hits the wall.

b) The maximum force exerted on the bumper during impact.



Energy question B

1) A deep fat frier uses 2 kJ of electrical energy per second. It can raise the temperature of 1.5 kg of cooking oil from 20°C to 80°C in 6 minutes. If the specific heat capacity of the oil is 2400 J/kg, calculate the heat energy taken in by the oil every second and hence the efficiency of the frier.

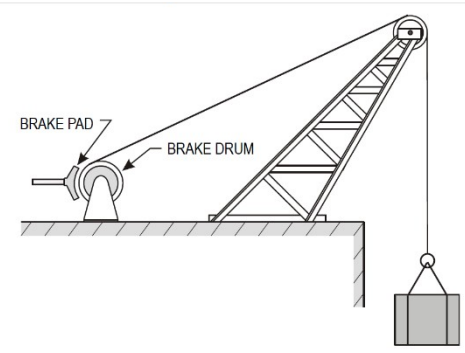
2) When full, a container of water holds 1.5 litres of water. The initial temperature of the cold water is 20°C. A heater operates from a 240 V supply, draws a current of 8 A and has an efficiency of 75%. Determine how long it will take to heat the container of water to 90°C.

3) A lift is used to raise a mass of 200 kg to a height of 10 m in 2 minutes. If the lift motor produces 12 MJ of energy in one hour, how efficient is the lift?



4) A hot drinks vending machine heats one cupful of water (0.15kg) from 30°C to 90°C. The heating element operates from a 240 volt supply and has a resistance of 12 ohms.

a) Calculate the current drawn by the heating element.
b) Calculate the heat energy transferred to the water.
c) Calculate the time taken to heat the water if the system is 100% efficient.
d) Calculate the percentage of energy lost if it actually takes 10 seconds to heat the water.
e) Draw a systems diagram showing the input and outputs to the system. Include the percentages.



5) A derrick is used to lower containers into the hold of a ship as shown. The speed at which the containers are lowered is controlled by a brake pad pushing against the side of a brake drum.

The derrick is used to lower 1 tonne containers at a steady speed.

a) Calculate the potential energy lost by the containers as they are lowered.

b) Assuming all this energy is converted into heat energy at the brake drum, use the following data to calculate the change in temperature of the drum.

Specific heat capacity of drum = 400 J/kgK and Mass of brake drum = 20 kg. When the containers are lifted out of the hold using a petrol driven hoist, 0.03 kg of petrol is used.

c) Calculate the amount of fuel energy supplied if the specific energy of petrol is 20 M J/kg.

7) A spring has a length of 500 mm when no external forces are present. When a force of 600 N is applied to the spring it doubles its length.

a) Calculate the spring stiffness in N/mm.

b) What size of force is required to increase the length to 600 mm.

c) Calculate the Work done in stretching the spring from 600 mm to 800 mm.

8) The spring in an air rifle has a stiffness of 5 N/mm and a free length of 120 mm. The spring is compressed half the distance just before a slug of mass 1.0 gram is fired from the rifle.

a) Calculate the energy stored in the spring when the slug is inserted.

b) What is the velocity of the slug immediately after it has been fired?

9) Cars are designed with a safety cage, which surrounds the passengers, and a crumple zone front and back to absorb any energy on impact. During a test run, a car of mass 800 kg is driven at 12 m/s into a solid wall. The crumple zone deforms but the safety cage remains intact thus protecting the passengers. The crumple zone deforms 500 mm on impact.

a) Calculate the kinetic energy of the car just before impact.

b) What will be the average stopping force (work) acting on the car during impact?

10) A piece of brass of mass 5 kg is dropped onto a hard surface without rebounding, resulting in a temperature rise of 2 °C. The specific heat capacity of brass is 370 J/kgK. Calculate the speed with which the brass hits the surface.